MeetingToM: Evaluating Multimodal LLMs on Theory-of-Mind Reasoning in Multi-Party Meetings

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Abstract

Theory of Mind (ToM), the ability to infer others' beliefs, intentions, and states of knowledge, is central to social interaction, yet remains challenging for current Multimodal Large Language Models (MLLMs), especially in multi-party meetings where cues are distributed across speech and behavior. Existing multimodal ToM benchmarks mainly focus on video-grounded question answering over overt, externally verifiable signals, and provide limited coverage of nuanced social states and group dynamics. We introduce **MeetingToM**, a benchmark for complex social behavior reasoning in naturalistic multi-party meetings. MeetingToM targets meeting-specific phenomena such as **pseudo-consensus**, where apparent agreement co-occurs with cues of dissent under social pressure. The benchmark is hierarchically organized to evaluate ToM at increasing levels of social granularity, including (i) subject-level mental state prediction, (ii) dyadic-level addressee understanding, and (iii) group-level consensus reasoning. We provide a unified evaluation protocol and conduct systematic analyses of representative MLLMs, revealing persistent limitations in integrating non-verbal cues, predicting annotated attitudes, and distinguishing among consensus states. Our results highlight key challenges and establish MeetingToM as a testbed for advancing meeting-grounded ToM in multimodal models. **Project page:** https://oliviaziyi.github.io/MeetingToM-Project-Page/

1 Introduction

Humans continuously infer and update their own and others' beliefs, intentions, and states of knowledge during social interactions. This capability, known as Theory of Mind (ToM), is fundamental to social behavior reasoning. Prior work shows that ToM is context-sensitive (Speiger et al., 2025), with goals, plans, and social norms shaping how people reason about others' belief states. Notably, these

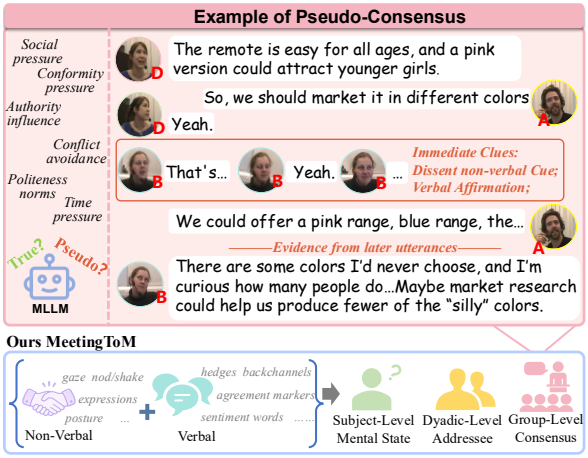


Figure 1: *Hierarchically structured tasks in our MeetingToM benchmark and example of pseudo-consensus.* MeetingToM evaluates meeting-grounded ToM reasoning at three levels: subject-level mental states, dyadic addressee identification and attitude inference, and group-level consensus reasoning. The example shows pseudo-consensus, where verbal agreement masks non-verbal dissent under social pressure.

contextual factors are especially salient in multi-party meetings, where participants bring distinct goals, partial knowledge, and social constraints. Accordingly, we seek to probe computational Theory of Mind in meeting scenarios. An AI system capable of interpreting complex social states in such a setting would enable practical applications such as collaborative decision support and socially aware assistants.

With the advent of Multimodal Large Language Models (MLLMs), there has been growing interest in probing their capabilities for Theory of Mind from a multimodal perspective, motivated by the fact that beliefs and intentions are often conveyed through both verbal and non-verbal cues. Existing multimodal ToM benchmarks (Jin et al., 2024a; Shi et al., 2025) largely emphasize video-grounded question answering, often focusing on overt, externally verifiable signals, rather than systematically

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testing nuanced, latent social states that unfold in realistic group interactions.

Multi-party meetings, however, present a particularly challenging and consequential setting for multimodal ToM. Beyond understanding what is said, models must piece together evidence from speech and behavior, including turn-taking, gaze, posture, and how people respond to each other (Pentland, 2008), while considering norms such as politeness and conformity. Crucially, meetings frequently contain surface-level agreement that masks private reservations: participants may verbally endorse proposals while subtly signaling doubt or disengagement, and disagreements may only become apparent through subsequent actions or indirect cues. Such phenomena call for benchmarks to explicitly evaluate whether models can infer hidden attitudes and group-level social dynamics, including cases where consensus is only apparent.

To this end, we introduce **MeetingToM**, a benchmark for evaluating MLLMs' capabilities for complex social behavior reasoning in naturalistic multi-party meetings. Consider the example in Fig. 1. We present a case in which participant B verbally agrees with another participant, yet their body language suggests otherwise, and subsequent utterances reveal underlying concerns. This social phenomenon, where apparent agreement masks private dissent under social pressure, is closely related to groupthink and conformity effects (Janis, 2008). In MeetingToM, we refer to this meeting-specific manifestation as **pseudo-consensus**. Concretely, our benchmark tasks are hierarchically organized, with increasing demands on ToM capabilities:

- **Subject-Level Mental State Prediction**: Infers a target participant's mental state from a brief video segment and aligned utterances.
- **Dyadic-Level Addressee Understanding**: Identifies the referent of "you" using gaze and body orientation cues, followed by inferring the addressee's attitude.
- **Group-Level Consensus Reasoning**: Determines whether group agreement is genuine, reflects pseudo-consensus, or is absent, and identifies any participants who privately dissent.

In our experiments, we first benchmark both proprietary and open-source MLLMs. We then analyze how input modalities, thinking prompts, discourse markers, and contextual priors affect meeting-grounded social-state understanding.

2 Related Work

Social Intelligence in Meeting Scenarios. Early research in social signal processing seeks to quantify group dynamics through observable behaviors such as exchange of conversational turns, recognition of roles, and establishment of dominance hierarchies (Vinciarelli et al., 2008; Pentland, 2008). Recent studies have transitioned toward modeling higher-level social constructs like group cohesion via nonverbal cues (Sanchez-Cortes et al., 2012; Lee et al., 2024). For example, (Vinciarelli et al., 2008) modeled social signaling through prosodic features, while (Jayagopi et al., 2009) utilized visual attention to infer dominance. Recently, (Ouyang et al., 2025) proposed a multi-speaker attention alignment framework to characterize social targets in group-level rapport. However, these approaches mostly focus on direct behavioral signals, without reasoning about inherent social dynamics, such as pseudo-consensus, that mask private disagreement under social pressure (Almaatouq et al., 2020; Villa-Cueva et al., 2025). Our benchmark bridges this gap by introducing novel tasks that identify both internal cognitive states and complex social dynamics in meeting scenarios.

Multimodal Social AI Benchmarks. Previous multimodal benchmarks primarily evaluate dyadic social affect recognition (Zadeh et al., 2018; Busso et al., 2008; Poria et al., 2017). Representative examples include Social-IQ (Zadeh et al., 2019), which targets in-the-wild multimodal social question answering, and SIV-Bench (Kong et al., 2025), which focuses on interaction reasoning over social scenes, states, and dynamics. The ToM-oriented resources such as MoMentS (Villa-Cueva et al., 2025) emphasize narrative-centric mental state inference, while MM-Soc (Jin et al., 2024b) targets socially situated online multimodal content. More recent benchmarks introduced in 2025 further push toward evidence-grounded and multi-agent ToM reasoning: Social Genome (Mathur et al., 2025) evaluates grounded social reasoning with human-annotated reasoning traces over multimodal interactions, and MuMA-ToM (Shi et al., 2025) tests embodied multi-agent Theory-of-Mind inference from video and textual contexts. Meeting corpora offer a distinct testbed for multiparty professional collaboration, yet large-scale resources such as MISP 2025 (Gao et al., 2025) are largely limited to audio-visual diarization. In contrast, our



Figure 2: *Overview of the hierarchical tasks in our MeetingToM benchmark.* Subject-level task assesses mental state prediction in meeting scenarios. Dyadic-level Task evaluates addressee identification and interpersonal attitude inference. Group-level tasks probe group consensus reasoning, including consensus quality assessment and the identification of hidden disagreement. These three tasks probe progressively richer social reasoning over verbal and non-verbal cues.

benchmark probes the in-depth reasoning ability of MLLMs, spanning subject-level internal states, dyadic-level reference understanding, and group-level consensus reasoning.

Theory of Mind in Multiparty Interaction. Theory of Mind (ToM) concerns inferring and tracking latent mental states, including beliefs, intentions, and false beliefs. Early NLP work evaluated ToM mainly via story-based QA (e.g., ToM QA Dataset (Nematzadeh et al., 2018)), but later analyses showed that common setups can be addressed by exploiting dataset regularities (Le et al., 2019). Recent LLM benchmarks for computational ToM primarily focus on dialogue-based QA, emphasizing higher-order belief recursion (Wu et al., 2023b), broader skill coverage (Chen et al., 2024), and long-form narratives with personality and intention cues (Xu et al., 2024). Broader evaluations with human comparison report notable gaps (Kosinski, 2024; Strachan et al., 2024) and motivate explicit belief tracking and robustness-oriented assessment (Sap et al., 2022; Sclar et al., 2023). Beyond text modality, ToM is increasingly tested in video-grounded QA (Jin et al., 2024a; Villa-Cueva et al., 2025), interactive dialogue requiring belief updates (Qiu et al., 2024; Yu et al., 2025), and role play or embodied multi-agent settings (Shinoda et al., 2025; Shi et al., 2025). Our work shares the motivation of multimodal ToM modeling, with a specific focus on meeting scenarios, where shared goals and task-driven participant roles provide priors that shape how MLLMs model participants' internal states.

3 MeetingToM Benchmark

3.1 Overview

We introduce **MeetingToM**, a multimodal benchmark designed to evaluate MLLMs' capability to reason about annotated social states in naturalistic multi-party meeting scenarios. Our benchmark is motivated by three core observations about real-world meetings: (1) participants continuously attribute their own and others' mental states; (2) referential expressions require joint modeling of discourse context, speaker intent, and participant roles for referent disambiguation; and (3) apparent group consensus can exert social pressure that masks individual disagreement. These phenomena demand integrated reasoning over visual cues such as facial expressions, gestures, and posture, as well as verbal content such as transcripts. Concretely, MeetingToM comprises three hierarchically structured tasks with progressively increasing Theory of Mind (ToM) reasoning depth:

- Task 1 targets *subject-level* mental state identification with multimodal signals.
- Task 2 requires *dyadic-level* understanding about

State	Closeup View	Brief Description	Non-verbal clues	Utterances
Active Engagement		*Actively driving the task , frequent initiative in speaking, and clear cognitive investment in the conversation.*	• Expressive facial expression	"We are thinking along those lines, just a different shape."
Supportive Endorsement		*Signalling agreement of others' ideas through brief affirmations and light, alignment-oriented elaborations.*	• Repeated nodding • Approving expression	" yep, yep..."
Focused Listening		*Closely attending to others' talk, while selectively processing and mentally integrating the information.*	• Focused expression • Gaze on the speaker	No verbal signals
Cognitive Conflict		*Challenging proposals that clash with one's knowledge, values, or goals, while remaining oriented to the task.*	• Rapid gaze shifts • Pointing at others	"Is it everybody is going to evaluate , or just the Market?"
Confused Bewilderment		*Experiencing cognitive disequilibrium due to overload or ambiguity, leading to partial loss of understanding.*	• Blank expression • Furrowed brows	No verbal signals
Hesitation		*A shortlived state between understanding and action, marked by disfluency or gaze aversion before speaking.*	• Hesitant expression • Scratching his head	"Probably...required to um... mo...most of the displays you see..."
Disengaged Withdrawal		*Pulling back from the meeting, with face compliance, behavioural signs of impatience or restlessness.*	• Distracted Focus • Frozen posture	No verbal signals

Figure 3: *Illustration of participant cognitive state definitions in meeting scenarios.* For each state, we show a representative close-up view, a brief semantic description, characteristic non-verbal cues (e.g., facial expressions, gaze, and gestures), and example utterances when present. These examples highlight how fine-grained mental states manifest through both verbal and non-verbal cues.

referential intent and interpersonal attitudes.

- Task 3 demands *group-level* reasoning about consensus and pseudo-consensus.

In the following sections, we first formally define the MeetingToM benchmark tasks and then detail the data sources and annotation process.

3.2 Task Definition

Input Notations. We formalize MeetingToM as a multiple-choice visual question answering benchmark that requires models to select the correct option based on video inputs captured from specific perspectives and accompanying conversational utterances. We denote conversation utterances as $\mathcal{U}$ and the video segment from view p as $\mathcal{V}^p$, where $p \in \{g, A, B, C, D\}$, where g denotes the global view and A–D denote the close-up views of the four participants, respectively. We also define a mosaic view $\mathcal{V}^m$ by spatially concatenating the synchronized closeup views $\{\mathcal{V}^A, \mathcal{V}^B, \mathcal{V}^C, \mathcal{V}^D\}$ into a 2×2 four-panel video.

Task 1: Subject-level Mental State Prediction. This task evaluates whether models can predict each participant's annotated state within a short temporal window, given the corresponding video segment and utterances. Building on prior work in psychology and cognitive science (Apperly, 2012), we define the following meeting mental states as annotation labels:

- ACTIVE ENGAGEMENT: Actively leading or driving the discussion (e.g., Kahn, 1990)
- SUPPORTIVE ENDORSEMENT: Showing agreement or validation (e.g., Edmondson, 1999)
- FOCUSED LISTENING: Attentive but neutral reception (e.g., Broadbent, 1958)
- COGNITIVE CONFLICT: Experiencing disagreement or skepticism (e.g., De Dreu, 2006)
- CONFUSED BEWILDERMENT: Exhibiting confusion or uncertainty (e.g., Sweller, 1988)
- HESITATION: Pausing before responding or acting (e.g., Clark and Fox Tree, 2002)
- DISENGAGED WITHDRAWAL: Showing low engagement or attention drift (e.g., Kahn, 1990)

Fig. 3 presents a visual illustration of these mental states, and detailed definitions are included in the Appendix A.

Given a short close-up video clip of the target subject A, denoted by $\mathcal{V}^A$, and the utterances $\mathcal{U}$, the model predicts A's annotated state:

$$\hat{s}_A = \arg\max_{s \in \mathcal{S}} p\big(s \mid \mathcal{V}^{\mathrm{A}}, \mathcal{U}, \mathcal{Q}^{\mathrm{ms}}\big), \qquad (1)$$

where $\mathcal{S}$ is the set of aforementioned mental states, and Q^{ms} is the mental state probing query. This task uses verbal and nonverbal evidence, including speech, facial expressions, and gaze.

Task 2: Dyadic-Level Referential Reasoning. This task considers two aspects of pairwise social behavior reasoning based on specified visual views. When a speaker A uses the pronoun "you", the model performs **(i) Addressee Identification** to determine who is being addressed and **(ii) Attitude Inference** to assess the attitude the addressee holds toward A's statement.

Given the global view $\mathcal{V}^{\mathrm{g}}$ and surrounding utterances $\mathcal{U}$, and Addressee Identification query $\mathcal{Q}^{\mathrm{addr}}$,

this task is defined as

$$\hat{r} = \arg\max_{r \in \mathcal{R}} p\big(r \mid \mathcal{V}^{\mathrm{g}}, \mathcal{U}, \mathcal{Q}^{\mathrm{addr}}\big), \quad (2)$$

where $\mathcal{R}$ is the addressee label space including: A, B, C, D, MULTIPLE, and UNKNOWN.

Correspondingly, the Attitude Inference task is defined as follows:

$$\hat{m}_{\hat{r} \to A} = \arg\max_{m \in \mathcal{M}} p\big(m \mid \mathcal{V}^{\mathrm{m}}, \mathcal{U}, \mathcal{Q}^{\mathrm{att}}, \hat{r}\big). \quad (3)$$

where $\mathcal{M}$ is the attitude label space with four classes: SUPPORT, OPPOSE, NEUTRAL, and UNCERTAIN. $\hat{r} \to A$ denotes the addressee's attitude toward speaker A's statement, while $\mathcal{Q}_{\mathrm{att}}$ denotes the attitude inference prompt.

These two tasks are particularly challenging. The Addressee Identification task requires models to reason about gaze direction and body orientation grounded in the specified pronoun and aligned utterances. The Attitude Inference task asks models to predict an annotated stance from verbal and nonverbal cues.

Task 3: Group-Level Consensus Reasoning. This task asks MLLMs to classify adjudicated consensus states, including pseudo-consensus. Specifically, at the group decision-making moment $t^{\star}$, the model performs **(i) Consensus Classification** to assess the quality of agreement and **(ii) Dissent Identification** to identify the dissenting participant when pseudo-consensus is detected, based on the mosaic view $\mathcal{V}^{\mathrm{m}}$ and the utterances $\mathcal{U}$. Consensus Classification is defined as

$$\hat{c} = \arg\max_{c \in \mathcal{C}} p(c \mid \mathcal{V}^{\mathrm{m}}, \mathcal{U}, \mathcal{Q}^{\mathrm{cons}}), \quad (4)$$

where $\mathcal{C}$ is the consensus label space with four classes: TC (True Consensus), PC (Pseudo Consensus), NC (No Consensus), and U (Uncertain).

If $\hat{c} =$ PC, the model identifies the hidden dissenter with dissent identification query $\mathcal{Q}^{\mathrm{diss}}$:

$$\hat{n} = \arg\max_{n \in \mathcal{N}} p\big(n \mid \mathcal{V}^{\mathrm{m}}, \mathcal{U}, \mathcal{Q}^{\mathrm{diss}}\big), \quad (5)$$

where $\mathcal{N}$ denotes the set of participants, with NONE indicating no dissenter is presented.

This task probes into an interesting social phenomenon: in professional settings, social pressure often leads individuals to verbally agree while their nonverbal behavior reveals reservation or disagreement. Detecting such pseudo-consensus requires identifying subtle incongruence between verbal affirmation and dissent visual cues such as crossed arms, delayed responses, or gaze aversion.

3.3 Data and Annotation

Source Data. MeetingToM is built on the AMI Meeting Corpus (McCowan et al., 2005), which provides synchronized multi-view meeting videos, word-level transcripts, and metadata. We use scenario-based product-design meetings, where four participants discuss proposals and decisions, yielding natural cases of agreement, disagreement, and consensus formation.

Video Clip Extraction. We extract task-specific clips around socially relevant events. Task 1 uses 5-second close-up clips of individual participants, while Tasks 2 and 3 use approximately 50-second clips to retain context for referential resolution and consensus reasoning. For each span, we synchronously cut the relevant video views, audio, and transcript using identical timestamps; inputs include global, close-up, or mosaic views depending on the task. Benchmark files and reconstruction resources are publicly available; see Appendix A.6.

Annotation Pipeline. We use a hybrid pipeline combining task-driven candidate extraction, GPT-assisted question generation, and human answer annotation:

1. **Candidate Identification**: We identify interaction segments with second-person pronoun usages, and decision-related dialogue patterns.
2. **Question Generation**: GPT-5 drafts candidate questions and task-specific options, but does not determine gold answers. Generated candidates were manually screened for answerability and sufficient evidence.
3. **Human Annotation**: Two annotators independently label each instance following task-specific preparation and calibration, with a third annotator resolving disagreements. Before production, all three annotators labeled about 20 shared pilot instances to calibrate labels and evidence criteria.
4. **Adjudication and Filtering**: Disagreements are adjudicated by a third annotator. Cases that remain ambiguous after adjudication are discarded, while retained disagreement cases are further reviewed and verified through joint discussion.

Original AMI annotations and metadata are used only for construction and validation. At evaluation time, models receive only the visual input, aligned transcript, and question; gold labels, AMI annotations, and auxiliary metadata are excluded. Further

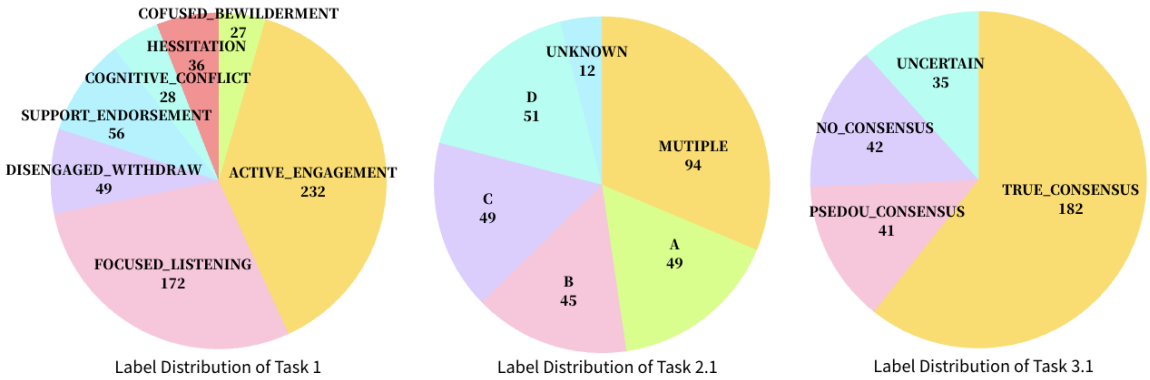


Figure 4: *Per-task label distribution*. We present the label distributions for representative tasks (1, 2.1, and 3.1).

details are provided in Appendix A.

Annotation Quality. We compute Fleiss' kappa (Fleiss, 1971) on an agreement subset of 200 instances per task, where all three annotators independently label the same items before consolidation. Task 1 achieves $\kappa = 0.7104$; Task 2 achieves $\kappa = 0.7319$ for addressee identification and $\kappa = 0.5014$ for attitude inference; Task 3 achieves $\kappa = 0.5681$ for consensus classification and $\kappa = 0.5729$ for dissenter identification. Lower agreement on attitude and consensus reflects greater ambiguity in these labels, with adjudication providing consistent reference labels for scoring.

Benchmark Statistics. MeetingToM contains 1,800 question-level items from 60 meeting sessions, with 600 items per task family. Figure 4 shows representative label distributions: Task 1 is skewed toward active and attentive states, Task 2.1 is more balanced, and Task 3.1 is dominated by true consensus. We therefore report macro-averaged metrics for imbalanced subtasks.

4 Experimental Results

4.1 Experimental Setup

Evaluated Models. We evaluate both proprietary and open-source MLLMs, including GPT-4o (OpenAI, 2024), GPT-5 (OpenAI, 2025), Gemini-3 Pro (Pichai et al., 2025), and Qwen2.5/3-VL models (Bai et al., 2025; Qwen-Team, 2025). For video-capable MLLMs, we provide the video clip, aligned transcript, and task prompt. For GPT-5 in our setup, each clip is converted into uniformly sampled frames at 0.5 FPS and paired with the same transcript and prompt. Additional input-formatting, video-ingestion, and prompt-template details are provided in Appendices A.5 and E.1.

Model	Task 1		Task 2.1	Task 2.2	Task 3.1		Task 3.2
	Acc.	Macro	Acc.	Acc.	Acc.	Macro	Acc.
chance-level	14.29	14.29	16.67	25.00	25.00	25.00	–
Protocol-informed ref.	86.33	75.79	86.00	83.33	80.33	74.97	80.00
Proprietary models							
Gemini-3 Pro	59.00	30.39	74.33	51.67	41.00	27.88	88.62
GPT-4o	36.06	14.23	29.10	46.15	21.28	10.13	58.73
GPT-5	55.67	28.69	54.33	51.00	43.33	36.57	90.77
Open-source models							
Qwen2.5-vl-32B	55.50	26.57	37.00	47.67	18.33	12.52	52.73
Qwen3-vl-8B	42.17	23.74	37.67	47.67	23.67	14.15	66.20
Qwen3-vl-32B	38.00	19.89	64.67	48.67	27.67	16.30	74.70

Table 1: *Overall performance of prevailing MLLMs on our MeetingToM benchmark.* The protocol-informed human reference remains higher across tasks, while proprietary models generally outperform open-source models. Macro is reported for imbalanced tasks.

Methods	Task1	Task2.1	Task2.2	Task3.1	Task3.2
Video	48.17	57.33	44.67	38.67	82.76
Utter.	61.33	72.67	55.00	55.67	97.60
Video & Utter.	59.00	74.33	51.67	41.00	88.62

Table 2: *Ablation of input modalities using Gemini-3 Pro.* Visual and verbal cues contribute differently across tasks, showing task-dependent effects of input.

Evaluation Metrics. We report exact-match accuracy for all tasks, together with a chance baseline under uniform random guessing for each subtask. Task 3.2 is evaluated conditionally on correctly answering Task 3.1, since dissenter identification is only meaningful when pseudo-consensus is detected. Since Tasks 1 and 3.1 exhibit substantial class imbalance, we also include Macro-F1 for these two tasks in Table 1.

4.2 Main Results

Overall Performance. The performance of both proprietary and open-source models is reported in Table 1. For comparison, the protocol-informed human reference remains above all evaluated MLLMs. Among proprietary models, Gemini-3 Pro performs best on Task 1 and the two dyadic subtasks, while

Model	Prompt	Task 1		Task 2.1	Task 2.2	Task 3.1		Task 3.2
		Acc.	Macro	Acc.	Acc.	Acc.	Macro	Acc.
Gemini-3 Pro	No Prompt	59.00	30.39	74.33	51.67	41.00	27.88	88.62
	Task-CoT	59.67 ↑0.67	27.48 ↓2.91	70.67 ↓3.66	56.00 ↑4.33	59.00 ↑18.00	28.52 ↑0.64	97.74 ↑9.12
	ECoT	61.33 ↑2.33	25.16 ↓5.23	74.00 ↓0.33	50.67 ↓1.00	55.33 ↑14.33	29.61 ↑1.73	97.59 ↑8.97
Qwen3-VL-32B	No Prompt	38.00	19.89	64.67	48.67	27.67	16.30	74.70
	Naive-CoT	42.17 ↑4.17	25.30 ↑5.41	66.00 ↑1.33	51.00 ↑2.33	33.00 ↑5.33	19.92 ↑3.62	79.80 ↑5.10
	Task-CoT	53.33 ↑15.33	27.47 ↑7.58	60.00 ↓4.67	49.67 ↑1.00	21.74 ↓5.93	12.39 ↓3.91	58.46 ↓16.24
	ECoT	54.33 ↑16.33	27.17 ↑7.28	57.33 ↓7.34	46.67 ↓2.00	37.67 ↑10.00	22.93 ↑6.64	82.30 ↑7.60

Table 3: Performance comparison across prompting strategies. Macro is reported for imbalanced tasks; arrows show absolute changes from No Prompt (↑ improvement, ↓ degradation). Task-CoT denotes task-aligned CoT.

GPT-5 achieves the strongest group-level results on Task 3.1 and Task 3.2. GPT-5 also substantially outperforms GPT-4o, especially on Task 1 and the group-level tasks. Open-source models remain competitive on selected subtasks, such as Qwen3-VL-32B on Task 2.1, but generally lag behind the strongest proprietary models across the full benchmark.

However, the leaderboard alone does not tell the full story. The additional Macro scores reveal clear class-imbalance effects: on Task 1, Gemini-3 Pro drops from 59.00 Acc. to 30.39 Macro, while GPT-5 drops from 55.67 to 28.69. This gap suggests that aggregate accuracy overstates model reliability by rewarding performance on frequent or visually salient states while masking failures on minority or context-sensitive states. A similar pattern appears in Task 3.1, where Gemini-3 Pro drops from 41.00 Acc. to 27.88 Macro and GPT-5 drops from 43.33 to 36.57. These gaps indicate uneven performance across consensus states, rather than uniformly reliable group-level reasoning.

Cross-Task Discussion. Consistent with the hierarchical design of MeetingToM, performance decreases as reasoning becomes more relational and collective. Subject-level mental-state recognition is comparatively easier, while dyadic and group-level tasks require aligning utterances, addressees, nonverbal behavior, and interaction context. For group-level reasoning, Task 3.1 is the main bottleneck; Task 3.2 is evaluated only when the consensus state is correctly identified.

Ablation on Input Modalities. We conduct input-modality ablations with Gemini-3 Pro and report the results in Table 2. Utterance-only input performs best on Tasks 1, 2.2, 3.1, and 3.2, suggesting that meeting-grounded social reasoning relies strongly on verbal content and conversational structure. This is especially important for attitude inference and consensus reasoning, where models must track what is said, whether a stance is explicit or implicit, and how participants converge or diverge. However, verbal input alone does not dominate all subtasks. For Task 2.1, combining video and utterances performs best, indicating that addressee identification benefits from aligning linguistic cues with gaze, body orientation, and visual engagement. The joint setting also does not consistently outperform utterance-only input on higher-level tasks, indicating that multimodal gains are task-dependent when cues are ambiguous. Overall, the modality results suggest that the challenge is not simply whether evidence is available, but whether models can combine it in a task-appropriate way.

Discourse Marker Shortcut Analysis. To test whether models rely on shallow transcript shortcuts, we remove hesitation and backchannel markers such as "hmm" and "emm" while keeping the visual input and task prompt unchanged. As shown in Table 5, removing these markers improves Task 1 but reduces performance on dyadic and group-level tasks. This suggests that such markers may sometimes distract local mental-state recognition, while still carrying useful interactional information for addressee, attitude, and consensus reasoning.

Benefits from Thinking Prompts. Thinking prompts can improve some VQA tasks. We test whether they help MeetingToM, where models reason over verbal, non-verbal, and interactional cues. We compare Chain-of-Thought (CoT) (Wei et al., 2022) and Emotional Chain-of-Thought (ECoT) (Li et al., 2024). For CoT, we report two variants: Naive-CoT, which uses generic step-by-step reasoning, and Task-aligned CoT (Task-CoT), which adds task-specific guidance for organizing subject-, dyadic-, and group-level social evidence. Prompt templates are provided in Appendix E.1.

As shown in Table 3, prompting effects are task-

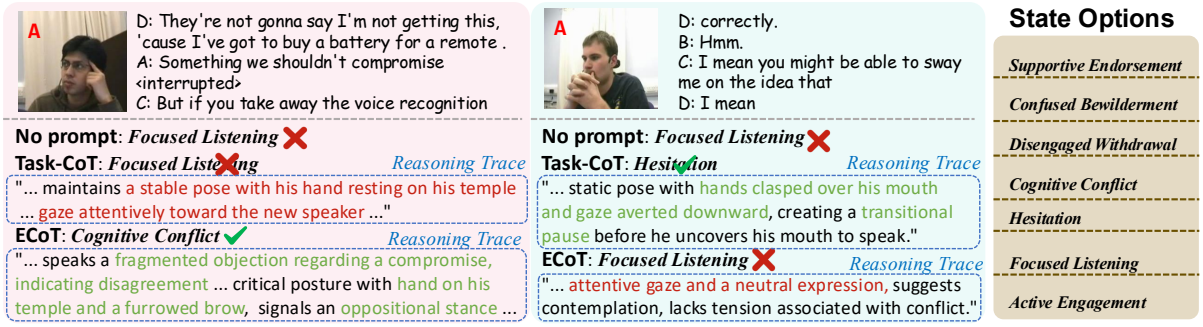


Figure 5: *Qualitative examples of prompting effects on subject-level mental state prediction.* The reasoning traces illustrate how ECoT and Task-CoT redirect attention to different social cues.

Methods	Task2.1	Task2.2	Task3.1	Task3.2
Qwen3-VL-32B	64.67	48.67	27.67	74.70
Base Model w. DA	35.67	43.00	24.33	73.97
Base Model w. phase	61.67	49.67	21.00	63.49
Base Model w. role	51.67	51.33	24.67	67.57

Table 4: Contextual-prior ablation results on Qwen3-VL-32B. Task 1 is not evaluated in this setting (n=0).

Setting	Task 1	Task 2.1	Task 2.2	Task 3.1	Task 3.2
Baseline	38.00	64.67	48.67	27.67	74.70
w/o discourse markers	52.50	60.00	46.33	21.67	63.08

Table 5: Discourse-marker ablation on Qwen3-VL-32B.

and model-dependent rather than uniformly beneficial. On Qwen3-VL-32B, Naive-CoT improves all subtasks, while ECoT improves Task 1 and group-level reasoning but reduces both dyadic subtasks. Task-CoT shows a different trade-off: it substantially improves Task 1, but reduces Task 2.1 and group-level performance. Gemini-3 Pro exhibits another pattern, with Task-CoT and ECoT mainly benefiting group-level tasks while producing mixed effects on Task 1 and Task 2. Additional open-source results in Appendix B.1 show that this lack of uniform gains is not specific to Qwen3-VL-32B. These results suggest that thinking prompts do not simply add reasoning depth; they reshape how models allocate attention across social evidence.

Fig. 5 illustrates this mechanism qualitatively. ECoT can direct the model toward expression-related cues and improve some subject-level predictions, but it may also over-emphasize facial affect while missing broader interactional evidence. Overall, prompting helps selectively, but does not solve the harder problem of calibrating verbal, visual, and interactional cues under social ambiguity.

Ablations on Contextual Priors. We further test whether structured priors improve Qwen3-VL-32B by adding participant role, meeting phase, or dialogue-act (DA) cues. Role, phase, and DA respectively indicate each participant's functional position, the current meeting stage, and each utterance's communicative function, all taken from the original human annotations in the AMI Meeting Corpus. As shown in Table 4, these priors do not yield consistent gains: DA sharply hurts addressee reasoning and modestly reduces group-level performance, while phase and role provide only isolated improvements. This suggests that structured priors are not automatically useful simply because they are semantically relevant. In meeting-grounded ToM, contextual labels become helpful only when calibrated against local verbal and non-verbal evidence; otherwise, they may be redundant with the video-transcript input or bias the model toward stereotyped expectations. These results point to a deeper bottleneck: current MLLMs struggle not only with missing context, but also with deciding when contextual information should constrain, revise, or yield to situated multimodal evidence.

5 Conclusion

We present MeetingToM, a multimodal benchmark for evaluating theory of mind capabilities in naturalistic meeting scenarios. Our benchmark comprises three task families: mental state recognition from short video segments, dyadic social reasoning about referential intent and interpersonal attitudes, and group-level consensus assessment. Experiments with state-of-the-art MLLMs reveal substantial room for improvement, particularly in detecting subtle nonverbal cues and resolving conflicts between verbal and visual signals. We investigate the benefits of chain-of-thought–style prompting and analyze the pitfall of injecting contextual priors. We hope MeetingToM will advance meeting-grounded social reasoning.

Limitations

First, MeetingToM currently draws from English-language meetings, and thus does not measure MLLMs' performance in multilingual settings. We plan to expand the sources to multilingual meetings in future versions. Second, constrained by current MLLMs' video context limits and the difficulty of annotating long-form interactions, we evaluate models on 5-second and 50–70-second key meeting segments rather than full-length meetings. MeetingToM also focuses on structured meeting scenarios, which may limit generalization to informal, online, or hybrid settings; we discuss these scope and temporal limitations further in Appendix D. Third, we cannot re-contact the original meeting participants to obtain self-reported mental states, so some labels may differ from their private beliefs or intentions. The moderate agreement on attitude- and consensus-related annotations also suggests residual ambiguity, and adjudication does not eliminate plausible alternative interpretations. In addition, all annotators are native Chinese speakers from similar Chinese/East Asian cultural backgrounds and use English as a second language, which may affect the interpretation of some social cues. Our evaluation also focuses on MLLMs and does not include a non-LLM structured-cue baseline.

Ethical considerations

MeetingToM is built solely from the publicly available AMI Meeting Corpus, following its documented access and distribution procedures. We do not collect new recordings and do not attempt to re-identify individuals beyond the anonymization already provided by AMI. Our annotations focus on short-lived, episode-bound mental states that are grounded in observable verbal and non-verbal cues within each segment. We explicitly avoid labeling stable personal traits or sensitive attributes. Since private mental states are not directly observable and we cannot obtain participant self-reports, the labels should be interpreted as systematic third-person inferences from available evidence rather than definitive ground truth about participants' internal experiences. We provide an extended discussion of potential misuse, bias, and deployment boundaries in Appendix C.

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A Conceptual Background for Meeting Mental States

This section summarizes the conceptual background for the seven meeting-specific mental states used in MeetingToM. The taxonomy is designed to jointly capture affective valence, task engagement, and typical interactional behavior, drawing on work in engagement, cognitive load, affect, and meeting science. Table 9 provides a compact overview.

Active Engagement. Active Engagement is a dynamic cognitive and affective state in which participants invest substantial attentional and emotional resources in the collaborative process, experiencing the meeting as meaningful and worth their effort. It aligns with psychological engagement (Kahn, 1990), self-determination theory on intrinsic motivation and goal congruence (Deci and Ryan, 2000), and work on psychological safety and collective efficacy in teams (Edmondson, 1999; Bandura, 2000). In meetings, Active Engagement typically appears as sustained focus, initiative taking, elaborated contributions, and cooperative problem solving, consistent with human performance models and meeting effectiveness research (Wickens et al., 2021; Allen and Lehmann-Willenbrock, 2023; Allen et al., 2015).

Focused Listening. Focused Listening is a cognitively active but externally restrained state in which participants selectively attend to others' speech, integrate new information, and anticipate intentions without yet taking the floor. It is grounded in selective-attention and cognitive-load accounts of information processing (Broadbent, 1958; Sweller, 1988), as well as work on mentalizing and perspective taking in social cognition (Frith and Frith, 2006). Subtle feedback behaviors such as nodding, gaze alignment, or brief backchannels signal this form of engaged reception in meetings and help maintain task alignment and conversational efficiency (Tennant et al., 2023; Allen and Lehmann-Willenbrock, 2023).

Supportive Endorsement. Supportive Endorsement denotes a positively valenced socio-emotional state of agreement, validation, and affiliative reinforcement toward others' ideas. It arises under conditions of trust, psychological safety, and perceived goal alignment (Edmondson, 1999; Kahn, 1990), and is linked to the broadening effects of positive affect on cooperation (Weiss and Cropanzano, 1996). Cognitively, it reflects social appraisal focused on consistency and feasibility rather than threat (Scherer, 2001); behaviorally, it appears as explicit agreement, praise, and light co-construction that strengthen cohesion and accelerate consensus in meetings (Allen et al., 2015).

Cognitive Conflict. Cognitive Conflict is a negatively valenced yet high-engagement state in which participants actively challenge ideas that contradict their knowledge, values, or goals. Organizational research distinguishes such task conflict from relationship conflict and shows that, when managed well, it can improve decision quality and innovation (Thomas, 1992; De Dreu, 2006). From a cognitive perspective, conflict monitoring and error detection processes are activated (Botvinick et al., 2001), while appraisal processes evaluate threats to goals and self (Lazarus, 1991). In meetings, Cognitive Conflict manifests as counterarguments, requests for justification, and attempts to correct inconsistencies; under psychologically safe climates it can be productive, but it can also escalate into interpersonal strain if mishandled (Edmondson, 1999; Allen and Lehmann-Willenbrock, 2023).

Confused Bewilderment. Confused Bewilderment describes a state of cognitive disequilibrium arising when informational demands exceed prior

knowledge or when instructions are ambiguous. Cognitive load theory characterizes this overload as a mismatch between task complexity and cognitive resources (Sweller, 1988), while affective-events theory explains the accompanying uncertainty and mild negative affect at work (Weiss and Cropanzano, 1996). Studies of learning show that confusion can be adaptive, triggering metacognitive monitoring and help-seeking when properly scaffolded (D'Mello and Graesser, 2012). In meetings, this state often appears as furrowed brows, delays, or clarification questions, reflecting attentional bottlenecks in processing complex or poorly structured information (Broadbent, 1958; Allen and Lehmann-Willenbrock, 2023). Unlike hesitation, confusion primarily reflects difficulty in understanding the task content rather than uncertainty about expressing one's view.

Hesitation. Hesitation is a short-lived cognitive and emotional bridge between uncertainty and overt action, characterized by speech disfluencies, timing delays, and visible self-monitoring before speaking. Psycholinguistic work shows that fillers such as *uh* and *um* signal difficulties in planning and coordination between thought and speech (Clark and Fox Tree, 2002; Fox Tree and Clark, 1997). Appraisal theories interpret hesitation as reflecting real-time checks of adequacy and social-evaluative risk (Scherer, 2001; Lazarus, 1991). In meetings, hesitation often precedes key turns in decision making and is observable in micropauses, hedging, and gaze aversion, which interact with temporal interaction patterns over the course of a discussion (Lehmann-Willenbrock and Allen, 2018). Unlike confusion, hesitation can arise even when the content is understood, reflecting uncertainty about the social consequences of speaking up.

Disengaged Withdrawal. Disengaged Withdrawal is a psychologically detached, negatively valenced state in which participants reduce cognitive and emotional investment in the meeting. It builds on engagement and disengagement accounts (Kahn, 1990) and burnout research on chronic strain and cynicism (Maslach and Leiter, 1997). Cognitively, attention drifts from task content toward thoughts about time and external concerns; affectively, boredom and frustration dominate and are located in low-arousal negative regions of the affective circumplex (Russell, 1980). In meeting science, withdrawal is visible as silence, multitasking, or minimal eye contact and is associated with low participation energy and ineffective meeting structures (Allen and Lehmann-Willenbrock, 2023; Allen et al., 2015; Rogelberg, 2019; Weiss and Cropanzano, 1996).

A Benchmark Construction and Scope

A.1 Candidate Mining and Clip Construction

We construct MeetingToM through a deterministic, task-driven pipeline. For each meeting, we first mine candidate temporal spans for the three task families and then instantiate non-overlapping clips within each meeting. For the subject-level STATE task, we mine interaction-based windows in which a target participant is visibly involved as a speaker or listener. For the dyadic YOU task, we locate utterances containing second-person pronouns and retain cases whose addressees can be resolved from surrounding verbal and visual context. For the group-level CONSENSUS task, we use AMI decision timestamps as anchors and further identify proposal-response structures and agreement markers around decision moments.

Task 1 uses 5-second close-up clips centered on the target participant, while Tasks 2 and 3 use 50–70 second clips, preferring 50 seconds when sufficient context is available. For each selected segment, we synchronously cut the Corner view, all Closeup views, audio, and transcript using the same start and end timestamps with `ffmpeg`. Mosaic views are constructed from synchronized close-up views when required by the task. This procedure ensures temporal alignment between verbal and non-verbal evidence.

A.2 GPT-assisted Question Generation

We use `gpt-5-2025-08-07` through the OpenAI Responses API to generate candidate questions and task-specific options. We set `max_output_tokens=8000` and leave temperature, top-p, and random seed unspecified. Questions follow fixed task formats, with task-specific fields such as the target participant, trigger utterance, addressee candidates, or consensus moment filled into predefined templates. Each generated candidate includes a provisional answer, rationale, and evidence-coverage score. The evidence-coverage score is used to rank candidates for manual review.

For each task, the 400 highest-ranked candidates are manually reviewed, and 300 are retained. Candidates are excluded when the target participant is

not clearly visible, the evidence is ambiguous or insufficient, or the available camera views do not support the question. The generated question text is kept without manual rewriting. Human annotators assign and adjudicate the final reference labels from the video and transcript.

A.3 Annotation Protocol and Quality Control

The annotation team consisted of three annotators: two undergraduate students and one master's graduate, with backgrounds in management, computer vision, and machine learning. One annotator had about one year of research experience in HCI and cognition. Before annotation, all annotators completed about one week of task-specific preparation, reviewed the guidelines and relevant Theory-of-Mind and cognitive-science literature, and independently labeled a shared pilot set of about 20 instances.

Each instance was independently labeled by two annotators using the video clip and transcript. Labels were based on observable verbal and nonverbal evidence. A third annotator resolved disagreements. The team held weekly one-to-two-hour calibration meetings to discuss ambiguous cases and update the guidelines. No numerical qualification threshold was used. Instances with unresolved ambiguity, insufficient evidence, temporal misalignment, missing views, or information outside the clip were discarded.

All three annotators are native Chinese speakers from similar Chinese/East Asian cultural backgrounds, with English as a second language. This may affect interpretations of social cues such as gaze, politeness, engagement, disagreement, and conformity.

A.4 Leakage Control and Prompt Consistency

Original AMI annotations and metadata are used only for benchmark construction and validation, and gold labels are used only for scoring. At evaluation time, the default model input contains only the visual input, aligned transcript, and task question. AMI dialogue acts, participant roles, meeting phases, and other auxiliary annotations are excluded unless explicitly introduced in controlled ablation settings.

To minimize prompt-specific bias, all evaluation prompts are predefined from the task definitions and applied uniformly across models. We do not perform model-specific prompt tuning or post-hoc prompt optimization based on experimental results.

A.5 Model Input Formatting and Video Ingestion

Our evaluation uses two ingestion pathways depending on model capability. For MLLMs that natively support video inputs, we provide the video clips directly together with the aligned transcript and task prompt. All information required for evaluation is consolidated in task-specific `.jsonl` files.

For GPT-5, we convert each clip into a sequence of uniformly sampled frames at 0.5 FPS and provide these frames together with the same aligned transcript and task prompt. Across all models, we keep the transcript, task prompt, answer space, and scoring protocol identical.

A.6 Benchmark Release and Reconstruction

The public release includes benchmark JSONL files, reconstruction metadata and scripts, evaluation code, annotation guidelines, and question-generation and evaluation prompts, available on GitHub. Each evaluation record is paired with reconstruction metadata specifying the AMI session, source time window, required camera view, and participant–view mapping.

Users can reconstruct the corresponding task-specific media from a local copy of the AMI Meeting Corpus using the released metadata and scripts. Transcript-conditioned inputs can be aligned from the AMI manual annotations using the released session and time-window information. Benchmark annotations are also available on Hugging Face.

A.7 Protocol-Informed Human Reference

The protocol-informed human reference was produced by one researcher familiar with the benchmark protocol, who completed all 1,800 question-level items. For each item, the evaluator received the audiovisual clip, aligned transcript, speaker-to-face mapping, question, and answer options. The adjudicated reference label was hidden during evaluation.

The evaluator's responses were scored against the adjudicated labels using the same benchmark evaluation script and metrics as the model outputs. We report accuracy for all subtasks and macro-F1 for the imbalanced Tasks 1 and 3.1, following the evaluation protocol in Table 1. This score represents one protocol-informed human reference rather than average human performance or a universal upper bound.

B Additional Prompting Diagnostics

B.1 Additional Open-Source Model Results

The prompting analysis in the main text shows that thinking prompts do not provide uniform gains across tasks or models. To further examine this pattern beyond the models reported in Table 3, we evaluate three additional open-source MLLMs: InternVL2.5-26B, InternVL3.5-8B, and LLaVA-Video-7B-Qwen2. As shown in Table 6, prompting effects remain non-uniform across these models, further supporting the observation that such strategies are task- and model-dependent.

B.2 Confusion Matrices and Class-wise Error Analysis

We report row-normalized confusion matrices for GPT-5, Gemini-3 Pro, and Qwen3-VL-32B on Tasks 1 and 3.1, together with class supports and the dominant off-diagonal errors. Rows denote adjudicated reference labels and columns denote model predictions.

Task 1. For compactness, we use the following abbreviations: DW = Disengaged Withdrawal, HES = Hesitation, CB = Confused Bewilderment, CC = Cognitive Conflict, SE = Supportive Endorsement, FL = Focused Listening, and AE = Active Engagement.

On Task 1, all three models frequently map other mental states to FOCUSED LISTENING, particularly CONFUSED BEWILDERMENT, DISENGAGED WITHDRAWAL, and SUPPORTIVE ENDORSEMENT.

Task 3.1. On Task 3.1, the largest row-normalized error for GPT-5 is UNCERTAIN→TRUE CONSENSUS (17/35=48.6%). GPT-5 also confuses NO CONSENSUS→PSEUDO CONSENSUS (13/42=31.0%), TRUE CONSENSUS→PSEUDO CONSENSUS (50/182=27.5%), and PSEUDO CONSENSUS→TRUE CONSENSUS (11/41=26.8%). Thus, the errors are directional rather than uniform: uncertain cases are frequently mapped to a substantive consensus category, and the model has difficulty separating true consensus from pseudoconsensus.

The complete results show an even stronger pseudoconsensus bias for the other models. Gemini-3 Pro predicts PSEUDO CONSENSUS for 71.4% of NO CONSENSUS instances and 48.4% of TRUE CONSENSUS instances, while Qwen3-VL-32B predicts PSEUDO CONSENSUS for 97.6% of NO CONSENSUS instances and 71.4% of TRUE CONSENSUS instances. Neither Gemini-3 Pro nor Qwen3-VL-32B selects UNCERTAIN for any Task 3.1 instance.

C Ethical Considerations and Broader Impacts

MeetingToM is intended as an evaluation benchmark rather than a deployment framework for inferring people's hidden intentions or attitudes in real-world settings. The labels in MeetingToM should be interpreted as systematic third-person inferences from observable evidence under a fixed annotation protocol, not as definitive ground truth about participants' inner mental states.

This distinction is important because Theory of Mind concerns latent states such as beliefs, intentions, uncertainty, and attitudes, which are not directly observable and must be inferred from behavior, language, and context (Premack and Woodruff, 1978; Chuey and Gweon, 2025). Human social inference is also fallible: people often overestimate how visible their internal states are to others, known as the illusion of transparency (Gilovich et al., 1998), and even deception judgments are only modestly above chance in many real-time settings (Bond Jr and DePaulo, 2006). Thus, ambiguity is not merely an annotation artifact but an inherent property of social inference.

This concern is especially relevant for AI systems. Recent work shows that strong models can perform well on constrained ToM benchmarks, yet their capabilities remain uneven and can lag behind human performance (Strachan et al., 2024; Jones et al., 2024; Chen et al., 2024). Other analyses further argue that standard benchmarks may capture only limited forms of ToM-like behavior rather than robust, functional social understanding (Riemer et al., 2024). Accordingly, high benchmark accuracy should not be interpreted as reliable access to another person's private mental state.

MeetingToM may also reflect representational and social bias. Because it is built on structured professional meetings, it may privilege particular norms of turn-taking, disagreement, attentiveness, and affect display. Systems evaluated on this benchmark may therefore generalize poorly to informal conversations, cross-cultural interactions, or neurodiverse communication styles, where signals of hesitation, dissent, engagement, or alignment may be expressed differently.

Model	Prompt	Task 1	Task 2.1	Task 2.2	Task 3.1	Task 3.2
InternVL2.5-26B	No Prompt	30.67	10.33	48.33	17.33	25.00
	Task-CoT	42.17 ↑11.50	17.00 ↑6.67	47.33 ↓1.00	16.00 ↓1.33	41.67 ↑16.67
	ECoT	36.83 ↑6.17	23.33 ↑13.00	48.33 (±0.00)	16.33 ↓1.00	40.82 ↑15.82
InternVL3.5-8B	No Prompt	31.00	22.33	48.00	18.00	70.37
	Task-CoT	28.33 ↓2.67	15.33 ↓7.00	44.00 ↓4.00	15.67 ↓2.33	27.66 ↓42.71
	ECoT	31.83 ↑0.83	19.00 ↓3.33	37.67 ↓10.33	18.67 ↑0.67	46.43 ↓23.94
LLaVA-Video-7B-Qwen2	No Prompt	26.83	19.00	44.67	58.67	0.00
	Task-CoT	41.33 ↑14.50	22.00 ↑3.00	44.00 ↓0.67	54.00 ↓4.67	1.85 ↑1.85
	ECoT	29.67 ↑2.83	24.00 ↑5.00	44.33 ↓0.33	49.33 ↓9.33	3.38 ↑3.38

Table 6: Additional prompting results on open-source MLLMs. Colored arrows indicate absolute changes relative to No Prompt.

Table 7: Row-normalized confusion matrices for Task 1.

Model	Adjudicated reference \ Prediction	DW	HES	CB	CC	SE	FL	AE	(n)
GPT-5	DW	**8.2**	6.1	0.0	0.0	2.0	73.5	10.2	49
	HES	0.0	**38.9**	0.0	2.8	5.6	38.9	13.9	36
	CB	0.0	0.0	**0.0**	3.7	3.7	85.2	7.4	27
	CC	0.0	21.4	3.6	**0.0**	14.3	28.6	32.1	28
	SE	0.0	0.0	0.0	0.0	**17.9**	66.1	16.1	56
	FL	0.6	1.2	0.0	0.0	5.2	**91.9**	1.2	172
	AE	0.9	22.0	0.9	1.3	3.9	7.3	**63.8**	232
Gemini-3 Pro	DW	**26.5**	4.1	2.0	0.0	4.1	61.2	2.0	49
	HES	2.8	**30.6**	0.0	2.8	8.3	38.9	16.7	36
	CB	3.7	0.0	**0.0**	7.4	0.0	81.5	7.4	27
	CC	0.0	3.6	0.0	**7.1**	3.6	28.6	53.6	28
	SE	3.6	0.0	0.0	0.0	**23.2**	58.9	14.3	56
	FL	3.5	1.7	0.6	0.6	6.4	**84.3**	2.9	172
	AE	1.3	10.3	0.4	4.7	3.9	6.0	**73.3**	232
Qwen3-VL-32B	DW	**10.2**	8.2	2.0	0.0	0.0	77.6	2.0	49
	HES	5.6	**30.6**	2.8	2.8	0.0	50.0	8.3	36
	CB	3.7	0.0	**7.4**	0.0	0.0	88.9	0.0	27
	CC	10.7	3.6	17.9	**0.0**	0.0	50.0	17.9	28
	SE	0.0	0.0	0.0	0.0	**1.8**	92.9	5.4	56
	FL	5.2	2.9	0.6	0.0	0.0	**91.3**	0.0	172
	AE	1.3	10.3	4.3	1.3	0.0	60.3	**22.4**	232

Note. For one Gemini-3 Pro Task 1 instance, generation terminated with `finish_reason=length` and returned no prediction. The instance is included in the class support and counted as incorrect.

A related risk is normative labeling. Our task formulation operationalizes complex social behavior into a finite set of mental-state and consensus categories. This abstraction enables controlled evaluation, but it should not be mistaken for a claim that there is always a single correct interpretation of another person's internal state. MeetingToM is best understood as a structured probe of model behavior under a fixed annotation scheme, not as a definitive ontology of human mental states.

We also note the possibility of dual use and misuse. Similar methods could be repurposed for workplace surveillance, automated assessment of employee engagement, or speculative judgment of disagreement, confidence, or attitude in professional settings. Such uses could disadvantage individuals whose communication styles differ from dominant norms or encourage overconfident inferences from incomplete evidence. We do not view MeetingToM as supporting high-stakes decision-making about real people, and benchmark performance should not be interpreted as evidence that such systems are ready for deployment in employment, educational, or managerial contexts.

Finally, prompting methods such as CoT or ECoT may improve benchmark performance while also encouraging fluent rationalizations for socially sensitive judgments. Such explanations can appear coherent even when weakly grounded in evidence. For this reason, prompting gains should not be read as evidence of calibrated, transparent, or unbiased social understanding.

Table 8: Row-normalized confusion matrices for Task 3.1.

Model	Adjudicated reference \ Prediction	NO	PSEUDO	TRUE	UNC	(n)
GPT-5	NO	**50.0**	31.0	14.3	4.8	42
	PSEUDO	26.8	**43.9**	26.8	2.4	41
	TRUE	23.1	27.5	**46.2**	3.3	182
	UNC	14.3	17.1	48.6	**20.0**	35
Gemini-3 Pro	NO	**19.0**	71.4	9.5	0.0	42
	PSEUDO	4.9	**75.6**	19.5	0.0	41
	TRUE	5.5	48.4	**46.2**	0.0	182
	UNC	8.6	20.0	71.4	**0.0**	35
Qwen3-VL-32B	NO	**0.0**	97.6	2.4	0.0	42
	PSEUDO	0.0	**87.8**	12.2	0.0	41
	TRUE	2.7	71.4	**25.8**	0.0	182
	UNC	11.4	62.9	25.7	**0.0**	35

Overall, MeetingToM should primarily be used to study model limitations, multimodal grounding, and evaluation methodology. Results should be interpreted together with task scope, class imbalance, annotation subjectivity, and domain specificity.

D Ecological Scope, Temporal Context, and Modern Meeting Settings

MeetingToM is built on the AMI Meeting Corpus. AMI contains in-person scenario meetings with synchronized multi-view video, transcripts, and relatively clear interaction structures. This setting provides a useful basis for constructing social reasoning tasks in meetings: participants share the same physical space, their roles and meeting goals are relatively explicit, and cues such as speech, posture, gaze, body orientation, and turn-taking can be observed in a relatively complete way. Therefore, MeetingToM is best understood as a controlled benchmark for evaluating latent social-state reasoning under meeting conditions where multimodal cues are rich and the interaction structure is relatively clear.

MeetingToM focuses on structured meeting scenarios because they provide rich multimodal signals, clear interaction structure, and practical relevance. This choice improves controllability and annotation consistency, but may limit generalization to other social settings, such as informal conversations or online and hybrid meetings. Our current design uses task-centered segments rather than full meetings to balance annotation reliability, precise reference-label alignment, and current MLLM video-context limits. This does not reduce the benchmark to local cue detection: Tasks 2 and 3 still require multi-turn context, including preceding proposals, responses, gaze shifts, and later clarifications. Extending the benchmark to longer meeting-level dynamics remains an important direction for future work.

Recent meeting benchmarks further clarify this positioning. QMSum, MeetingBank, and VCSum mainly focus on meeting summarization (Zhong et al., 2021; Hu et al., 2023; Wu et al., 2023a); MUG covers general meeting understanding and generation tasks such as topic segmentation, summarization, keyphrase extraction, and action item detection (Zhang et al., 2023); M2MeT focuses on multi-channel multi-party meeting transcription, including speaker diarization and multi-speaker ASR (Yu et al., 2022); MISP-Meeting / MISP 2025 support multimodal Mandarin meeting transcription, summarization, audio-visual speaker diarization, speech recognition, and joint diarization-recognition (Chen et al., 2025; Gao et al., 2025); and ELITR-Bench further moves toward meeting assistant evaluation over long noisy ASR transcripts (Thonet et al., 2025). These works have advanced models' ability to process meeting content, identify speakers, generate summaries, extract action items, and support meeting assistants. Overall, recent meeting benchmarks mainly treat meetings as information records or workflow objects: what was said, who said it, what should be summarized, and what tasks should be carried out.

However, real meetings are not only information exchange processes. They are also processes of social collaboration and group coordination. Participants do not always express their true stance directly. Due to politeness, power relations, group pressure, conflict avoidance, or time pressure, they may manage disagreement and consensus through

silence, brief agreement, hesitation, softened expressions, or delayed responses. For AI systems intended to operate in real social settings, recognizing meeting content alone is not enough. A socially aware system also needs to understand how participants respond to one another, who only appears to agree, who may still hold reservations, whether a decision has genuinely gained group support, and whether smooth visible progress masks hidden disagreement. In this sense, social reasoning is not a secondary issue in meeting understanding, but a core capability for AI systems that aim to participate in collaborative, organizational, and social interaction settings.

MeetingToM is designed to address this gap. It treats meetings as socially situated interactions rather than simply as information carriers. It focuses not only on what was said, but also on how participants display uncertainty, support, reservation, disagreement, weak buy-in, or pseudo-consensus during interaction. In this sense, MeetingToM complements recent meeting benchmarks by systematically evaluating multimodal social-pragmatic reasoning over latent interactional states.

This positioning also helps clarify the relation between AMI and modern remote or hybrid meetings. Meetings in AMI differ from contemporary platform-mediated meetings. In remote or hybrid meetings, camera placement, participant visibility, gaze interpretation, turn-taking, and participation patterns may all change. Yet what changes is not whether these social phenomena exist, but the cue system through which they are observed, interpreted, and annotated. Individual uncertainty, directed stance, weak buy-in, hidden dissent, and apparent consensus can still arise in remote and hybrid meetings. In in-person meetings, these states may be expressed through facial expression, gaze, posture, gesture, body orientation, turn-taking, and shared physical space. In video-mediated meetings, they may shift toward platform-mediated cues, such as camera status, mute/unmute behavior, response latency, chat feedback, emoji reactions, screen sharing, comments on shared documents, and differences in the meeting layout visible to different participants.

This shift affects the three levels of MeetingToM in different ways. For subject-level reasoning, hesitation, confusion, or disengagement may no longer be expressed mainly through posture or gesture, but through pauses, hedging, self-repairs, delayed responses, brief acknowledgments, camera-off silence, or chat-based participation. For dyadic-level reasoning, gaze and body orientation become less stable evidence for addressee relations. Models may need to rely more on explicit names, quoted references, reply structure, adjacent turns, chat replies, or actions during screen sharing. For group-level reasoning, pseudo-consensus may still occur, but it may appear through silence, minimal verbal agreement, ignored chat objections, unresolved comments on shared documents, or asymmetric participation between in-room and remote attendees. Hybrid meetings are especially complex because some alignment cues may be locally visible only to participants in the same room, while remaining invisible or delayed for remote participants.

These differences also imply that extending MeetingToM to remote and hybrid meetings would require more than simply replacing AMI with a newer corpus. It would require new annotation schemes for platform-mediated cues, uneven participant visibility, artifact-centered interaction, and asymmetric participation structures. For example, camera-off participation may indicate disengagement, privacy preference, bandwidth constraints, or platform norms depending on context; chat objections may be highly salient to some participants but invisible or ignored by others; and shared-document comments may function as delayed forms of disagreement or repair. These cues are not weaker versions of in-room signals, but different interactional resources that reshape how social states become observable.

Therefore, the value of MeetingToM lies in making the social reasoning dimension of meeting understanding explicit and systematic. Existing meeting benchmarks have advanced models' understanding of meeting content, speaker attribution, summarization, and action item extraction. Yet collaboration in real meetings is not composed only of explicit information; it also involves how participants respond to one another, adjust their stances, withhold disagreement, and form group consensus. AMI provides a research starting point where multimodal cues are relatively rich and the interaction structure is relatively clear. Extending this framework to remote and hybrid meetings is not only a matter of introducing newer meeting data, but also of understanding how platform-mediated meetings reshape the visibility, sharedness, and interpretability of social cues. In other words, modern meeting settings do not make social reasoning less relevant; rather, they make it appear through new cue forms.

Models need to connect language, vision, platform behavior, and interaction structure in order to better understand relations, stances, and group states in meetings.

E Mapping Meeting Mental States to AMI Annotations

E.1 Prompt Templates

E.2 Annotation Interface and Guidelines

Meeting mental state	Engagement	Valence	Expression	Key AMI-derived cues
Active Engagement	High	Positive	High	Dense *dialogueActs* involving informing, suggesting, and elaborating; frequent nods in *headGesture*; task-oriented *handGesture* such as pointing or writing; *focus* on the current speaker or shared artefacts; forward-leaning patterns in *movement* within the corresponding *segments*.
Focused Listening	High	Positive	Low	Long stretches with few floor-taking *dialogueActs* beyond backchannels; sustained *focus* on the current speaker; subtle nods in *headGesture*; minimal *handGesture* and low-amplitude *movement*, while the content of *words* stays on-topic.
Supportive Endorsement	Medium	Positive	Medium	Short agreement or positive-assessment *dialogueActs*; nodding *headGesture* aligned with others' turns; occasional affiliative *handGesture*; stable *focus* on the proposer during supportive comments; local *segments* that follow another participant's proposal in the same *topic*.
Confused Bewilderment	Medium	Negative	Low	Clarification or repair-initiation *dialogueActs*; increased *disfluency* in the *words*; tilts or brief freezes in *headGesture*; unstable *focus* alternating between speaker and materials; often appearing when *topics* shift or new information is introduced.
Hesitation	Medium	Negative	Low	Filled pauses and self-repairs in *disfluency*; delayed response after incoming *dialogueActs*; micro-shifts in *headGesture* and *focus*; small self-adaptive *handGesture* or posture changes within a single *segment*.
Cognitive Conflict	High	Negative	High	Disagreement or rejection *dialogueActs*; overlaps in turn-taking; head shakes in *headGesture*; expansive *handGesture*; rapid *focus* shifts between opponents and shared artefacts around contested *topics*.
Disengaged Withdrawal	Low	Negative	Low	Sparse *dialogueActs*; *focus* on table or objects rather than speakers; minimal task-related *handGesture*; leaning-back or slumped *movement*; off-topic or absent *words*.

Table 9: Meeting mental state taxonomy in MeetingToM. Engagement, valence, and expression describe coarse affective–behavioral profiles; the rightmost column shows how each state is operationalized using existing AMI-style annotation layers.

State prompt

[System]
"You are an evaluation model. Follow the output schema specified in the user prompt strictly.
[Guideline]
You will be given a short meeting clip video showing one target participant in upper body view only, and a transcript for the same time window. The target participant shown in the video is {target_agent}.
Base your decisions primarily on visual behavior in the video. The transcript is optional and may be used only to locate reaction moments and must not override what is visible.

Minimal label meanings:
ACTIVE_ENGAGEMENT means the target is actively contributing or leading, indicated by speaking with directive hand gestures, forward lean, animated expression, or active gaze engagement with others.
SUPPORTIVE_ENDORSEMENT means positive alignment or validation toward someone else's idea, indicated by repeated nodding aligned to others' speech, a warm approving smile, open posture, or quick affirmative reactions after key utterances.
FOCUSED_LISTENING means attentive but neutral receiving with no clear positive or negative stance, indicated by gaze mostly toward the speaker or screen, upright posture, and sparse small nods that are not clearly affirmative.
COGNITIVE_CONFLICT means disagreement rather than confusion, indicated by repeated rejecting movements, tight lips or tense jaw, defensive posture such as arms crossed or leaning back, dismissive gestures, or negative stance.
CONFUSED_BEWILDERMENT means cognitive overload rather than disagreement, indicated by puzzled brow or squint, head tilt, scanning others for clarification, mouth slightly open with a searching look, or self touch such as chin or forehead touch while thinking.
HESITATION means a transitional pause before speaking or acting with tentative turn taking, indicated by delayed response when a turn is available, mouth opens then closes, lip press then release, brief gaze aversion down or away then back, or a half gesture that starts then stops.
DISENGAGED_WITHDRAWAL means low engagement with attention drifting away from the interaction, indicated by gaze away down or to the side, leaning back with long stillness, fidgeting with objects, and a flat checked out affect.

Also provide a rationale (in 2 to 4 sentences) that references only observable cues.
The output format must strictly follow: {{"Q1":"<OPTION>","Q2":"<OPTION>","rationale":"<2-4 sentences>"}}.
[Context]
<target_corner_view.mp4, transcript.txt>

[Task-CoT]
Think step by step to decide the best label.

First assess whether the target is clearly visible, partially visible, poorly visible, or not visible at all. Then identify at least two clear cues across categories such as affect, gaze, posture, gesture, timing, and engagement. If you cannot identify clear cues, treat the evidence as insufficient and choose a conservative label.

Resolve common confusions as follows. Supportive endorsement requires clearly positive signals like a nod smile pattern; otherwise prefer focused listening. Confused bewilderment indicates comprehension difficulty signs; hesitation indicates turn taking pause or abort before speaking. Focused listening implies gaze mostly on the speaker with stable posture; disengaged withdrawal implies sustained gaze away or down with low reactivity.

[EoT]
Follow the Emotional Chain-of-Thought below and think step by step.

Step 1 Understanding context.
Infer what is happening in this moment and the target participant's likely role in the interaction. Use the transcript only to locate timing if needed.

Step 2 Recognizing emotions from visible cues.
From affect, gaze, posture, head movement, engagement level, gesture energy, and response timing, infer the target participant's likely emotional tendency, while tracking uncertainty when cues are weak.

Step 3 Recognizing self emotions.
Notice any immediate interpretive bias or emotional pull you feel while observing, and acknowledge uncertainty internally rather than forcing a confident story.

Step 4 Managing self emotions.
Regulate overconfidence and avoid projecting emotions onto the target. Keep an evidence-first stance and let uncertainty remain when the video does not support a clear inference.

Step 5 Influencing others' emotions and generating the final answer.
Synthesize the emotionally informed visual evidence into the single best option that matches the dominant state across the whole window. If the evidence is weak or mixed, choose the most conservative option and lower confidence.

Consensus prompt

[System]
"You are an evaluation model. Follow the output schema specified in the user prompt strictly.
[Guideline]
You will be given one meeting clip as a 2x2 closeup mosaic video. Both questions refer to the same attached mosaic video. Base your decisions primarily on visual behavior in the video. The transcript is optional and may be used only to locate reaction moments and must not override what is visible.

Minimal label meanings:
TRUE_CONSENSUS: broad, genuine alignment with no notable misalignment signals.
PSEUDO_CONSENSUS: surface agreement, but at least one participant shows persistent subtle misalignment (hesitation/tense/avoidant/skeptical), suggesting compliance rather than true buy-in.
NO_CONSENSUS: clear disagreement or mixed/fragmented signals across participants.
UNCERTAIN: insufficient visual evidence to distinguish the above.

Also provide a rationale (in 2 to 4 sentences) that references only observable cues.
The output format must strictly follow: {{"Q1":"<OPTION>","Q2":"<OPTION>","rationale":"<2-4 sentences>"}}.
[Context]
<mosaic_view.mp4, transcript.txt>

[Task-CoT]
Think step by step to decide Q1 and Q2.

First identify one or two clear reaction moments such as synchronized nods, affect shifts, or clear disagreement signals.

For Q1 choose the option that best matches the visuals. Remember label meanings mentioned above.
For Q2, if Q1 is PSEUDO_CONSENSUS and there is exactly one clearly standout subtle dissenter, then choose that agent. Otherwise if there are multiple candidates or ambiguous(you cannot reliably map from A,B,C and D), then Q2 must be NONE.

[EoT]
Follow the Emotional Chain-of-Thought below and think step by step.

Step 1 Understanding context.
Infer what is being discussed or decided in this moment and when the key reactions occur.

Step 2 Recognizing others' emotions.
From visible cues such as facial affect, gaze, posture, head movement, engagement, and timing, infer the likely emotional tendencies of each visible participant, while tracking uncertainty when cues are weak.

Step 3 Recognizing self emotions.
Notice your own immediate interpretive bias or emotional pull while observing the scene, and explicitly acknowledge uncertainty internally rather than forcing a confident story.

Step 4 Managing self emotions.
Regulate overconfidence and avoid projecting emotions onto participants. Keep a calm, evidence-first stance and let uncertainty remain when the video does not support a clear inference.

Step 5 Influencing others' emotions and generating the final answers.
Synthesize the emotionally informed visual evidence into the best Q1 label describing the group dynamic. For Q2, select a dissenter only if one participant stands out clearly; otherwise NONE.

You prompt

[System]
"You are an evaluation model. Follow the output schema specified in the user prompt strictly.
[Guideline]
You will be given two meeting clips from the same interaction window.
The first video is a corner view, which you must use to answer Q1 about addressee using layout, position, and orientation cues.
The second video is a 2x2 closeup mosaic, which you must use to answer Q2 about attitude using per-person facial and upper-body cues.
Answer primarily from visual evidence from the relevant video. Use the transcript only to help locate timing or turn-taking. If the transcript conflicts with what you see, trust the videos. Also provide a rationale (in 2 to 4 sentences) that references only observable cues, and may specify the originating video source (e.g., Corner or Mosaic).
The output format must strictly follow: {{"Q1":"<OPTION>","Q2":"<OPTION>","rationale":"<2-4 sentences>"}}.
[Context]
<2x2_corner_view.mp4, mosaic_view.mp4, transcript.txt>

[Task-CoT]
Think step by step to decide Q1 and Q2.

For Q1 focus only on the corner view during the addressing moment. Prefer consistent cues such as head and gaze orientation, pointing or directing hand gestures, body orientation and immediate visible reactions right after the addressing moment.

For Q2 focus only on the mosaic view and decide whose attitude is being asked about before judging attitude.
Supportive cues include nodding, smiling, warm approving affect, forward lean, and quick affirmative reactions.
Oppositional cues include head shakes, frowns, brow furrow, lip press, tense jaw, arms crossed, leaning back, dismissive gestures, and clearly negative stance.
Neutral cues include steady attentive gaze, stillness, neutral affect, and stable listening posture.

[EoT]
Follow the Emotional Chain-of-Thought below and think step by step.

Step 1 Understanding context.
Infer what is happening in this interaction window and when the key addressing moment occurs.

Step 2 Recognizing others' emotions and social signals.
From body orientation, positioning, approach/avoid movements, gaze direction when visible, and turn-taking patterns, infer who is being addressed in the moment, while tracking uncertainty when cues are weak.

Step 3 Recognizing self emotions.
Notice any immediate interpretive bias you feel while watching, and acknowledge uncertainty rather than forcing a confident story.

Step 4 Managing self emotions.
Regulate overconfidence and avoid projecting intentions onto participants. Keep an evidence-first stance and let uncertainty remain when the video does not support a clear inference.

Step 5 Influencing others' emotions and generating the final answers.
For Q1, decide the addressee based on the dominant addressing pattern in the corner view across the whole window.
For Q2, decide the attitude based on the dominant facial and upper-body affect patterns in the mosaic across the whole window.

Figure 6: MeetingToM's CoT and ECoT Prompt Examples

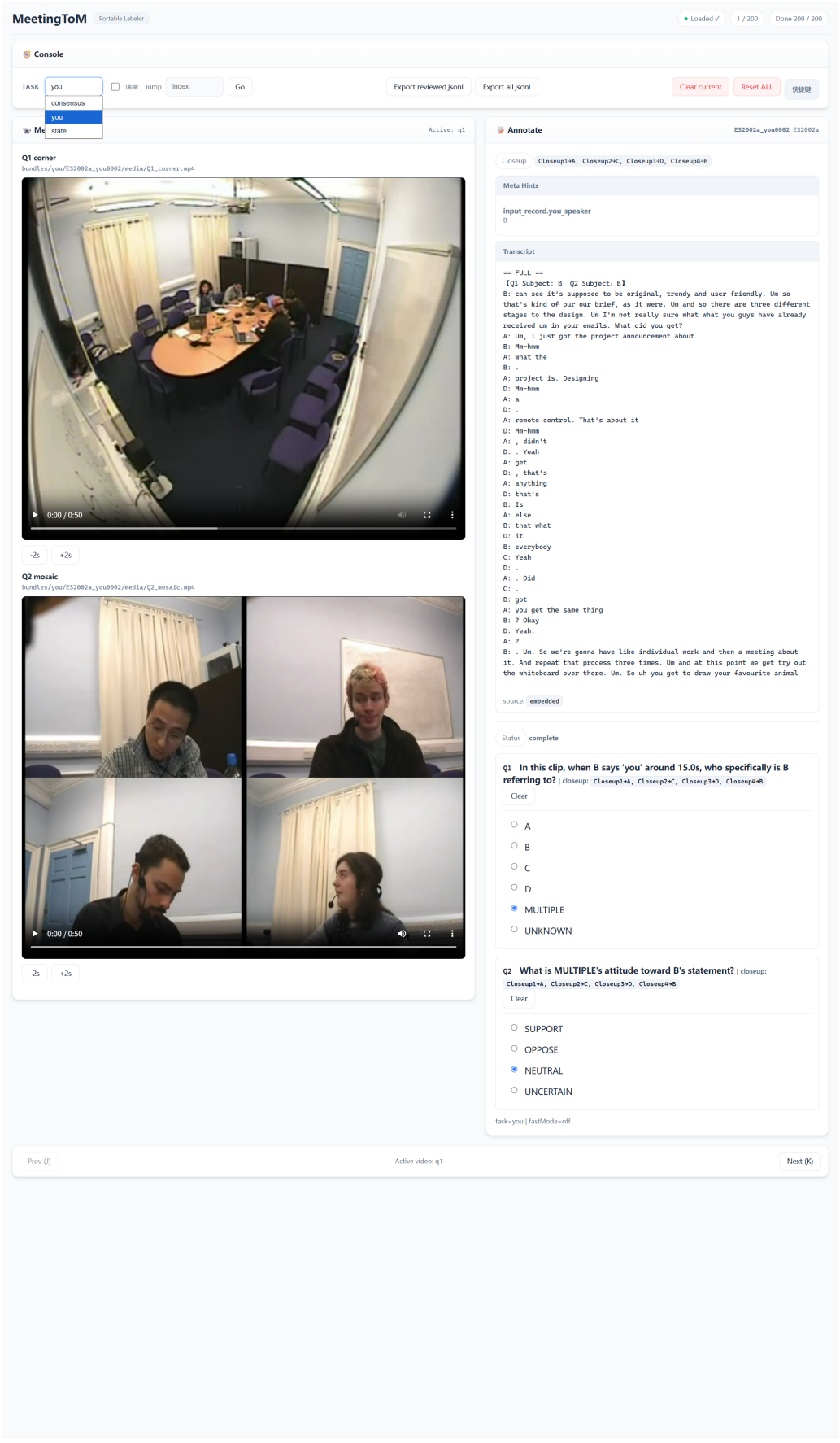


Figure 7: MeetingToM Annotation Page

Annotation Instructions (MeetingToM)

A. Inputs and Viewing Protocol

Annotators are given a ~50-second meeting clip from AMI with a corner (wide) view and participant close-ups ($A, B, C and D$), plus a transcript with timestamps.

Viewing protocol

- First pass (global context): watch the full clip once in corner view to understand topic flow and interaction structure.
- Second pass (evidence collection): re-watch focusing on the relevant time window (Task-dependent) using the target close-up as primary evidence, and the corner view as secondary evidence for gaze/turn-taking and group alignment.
- No over-interpretation: labels must be justified by observable visual cues; avoid inferring intentions that are not behaviorally supported.

Evidence priority

Visual cues (face, gaze, posture, gesture, timing) > transcript content > inference.
When cues conflict, prioritize stable, repeated, and temporally aligned signals over isolated micro-movements.

B. Task 1 Instructions: Participant Mental-State (Early vs Late)

Goal

For a specified target participant X, assign the participant's dominant mental/interaction state in two windows:

- Early window: 0–5 seconds
- Late window: last 5 seconds ($T - 5$ to T)

Step-by-step

Open the target close-up for participant X; use corner view to verify who is speaking and where attention is directed.
For each window, collect at least two supporting cues (e.g., gaze alignment + posture; facial affect + head movement; gesture + turn-taking role).
Assign one label per window from the Task 1 label set.
If the participant's state changes within a window, label the dominant state; if unclear, label the state most evident at the end of the window.

Key rules

- Label the target participant's state (not the speaker's, and not the group's).
- Use HESITATION only when "about-to-speak but holds back" behaviors dominate (e.g., lean-in then retreat; hand half-raise then drop; pre-speech pauses).
- If visibility is insufficient (face fully occluded/blurry for most of the window), mark as UNCERTAIN (or follow dataset policy for skipping).

C. Task 2 Instructions: "You" Addressee and Addressee Stance

When the speaker uses "you", annotate:

Q1. Addressee Identification

1. Step-by-step:
   Use transcript timestamps to locate the "you" token.
   Inspect the 1–2 seconds around the token in corner view and the speaker close-up.
2. Determine addressee using converging cues:
   Speaker gaze direction
   Pointing/indicating gesture
   Speaker torso orientation
   Which participant reacts (eye contact, nod, posture shift, verbal response)

3. Output one of: A/B/C/D/E, MULTIPLE (plural/group address), or UNKNOWN (insufficient evidence).

Q2. Addressee Stance Toward the Speaker

1. Time window: focus on the addressee's immediate reaction during and right after the "you" usage (typically ±2 seconds).
2. Step-by-step:
   Switch to the addressee close-up and check for stance cues (head movement, facial affect, posture openness/defensiveness, gaze engagement/avoidance).
3. Assign one stance label: SUPPORT / OPPOSE / NEUTRAL / UNCERTAIN.
4. Key rules ("politeness trap"):
   - If verbal content suggests agreement but body language shows resistance (e.g., arms crossed, lean back, stiff face), label based on visual stance.
   - NEUTRAL means attentive but non-committal (not disengaged).
   - Use UNCERTAIN if the addressee is not visible/too blurry or cues are strongly contradictory.

D. Task 3 Instructions: Group Consensus Quality (+ Hidden Dissenter)

Goal: At a decision/convergence moment, label the group consensus quality.

Q1. Consensus Quality

1. Step-by-step:
   Identify the decision/convergence moment (explicit proposal acceptance, "are we agreed?", summarizing decision, or clear topic closure).
   Use corner view to assess group-level alignment (shared attention, synchronized nods, posture openness, affect).
2. Assign one label:
   - TRUE_CONSENSUS: alignment signals across ≥2 participants and no salient dissent cues.
   - PSEUDO_CONSENSUS: surface agreement but one participant shows dissent "leakage" (e.g., stiff face, lip press, minimal/token nod, arms crossed, lean back, gaze away at commitment).
   - NO_CONSENSUS: overt disagreement or fragmented attention (head shakes, negating gestures, side-talk, widespread disengagement).
   - UNCERTAIN: key reactions not observable due to occlusion/quality or no interpretable convergence moment.

Q2. Hidden Dissenter (only if PSEUDO_CONSENSUS)

Locate the participant with the strongest and most temporally aligned dissent leakage at the agreement moment.
Output that participant's ID (A/B/C/D/E). Otherwise output NONE.

E. Quality Control Checklist (Annotator Self-Check)

Before submitting each item, confirm:

- I used the correct time window for the task (Task 1 early/late; Task 2 around "you"; Task 3 at convergence).
- My label is supported by ≥2 visual cues (unless the clip is clearly insufficient, then UNCERTAIN/UNKNOWN).
- I did not use later events in the clip to justify earlier-window labels.
- If cues were weak/contradictory, I chose the conservative option (NEUTRAL/FOCUSED_LISTENING/UNKNOWN/UNCERTAIN per task).

Figure 8: MeetingToM Annotation Guidelines